



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

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- Introduction
- Methodology
- Results
- Conclusion
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Executive Summary

- Summary of methodologies
 - Data collection with Web Scraping
 - Data wrangling
 - Exploratory Data using SQL
 - Exploratory Data Analysis for Visualization
 - Interactive Visual Analytics with Folium
 - Interactive Visual Analytics with Plotly Dash
 - Machine Learning Predictive Analysis for Classification
- Summary of all results
 - Exploratory Data Analysis Results
 - Interactive Analytics
 - Predictive Analysis Results

Introduction

Project Background and Objectives

This capstone project aims to understand the **key factors influencing the successful landing of SpaceX Falcon 9 rockets**.

Each launch costs more than **60 million USD**, yet the reusability of the first stage provides substantial cost savings compared to traditional expendable rockets.

The main objective is to develop a **predictive model** capable of accurately estimating the likelihood of a successful landing based on multiple mission parameters such as payload mass, orbit type, launch site, and booster version.

Such a model would not only help optimize launch planning and rocket design but also **enhance safety and operational efficiency**, contributing to more sustainable and cost-effective space missions.

Problem Statement and Research Question

This project seeks to answer a central question:

Given a set of features describing a Falcon 9 launch — including payload mass, orbit type, and launch site — will the rocket's first stage **successfully land**?

Answering this question requires a complete analytical process, involving **data collection, cleaning, exploration, visualization, and predictive modeling**.



Section 1

Methodology

Methodology

Executive Summary

A rigorous and structured methodology was adopted, consisting of the following key stages:

Data Collection:

Data were gathered through **web scraping** and the **SpaceX REST API**.

Data Wrangling and Preprocessing:

The raw data were cleaned and filtered. Missing values were replaced with mean values for numerical fields, and categorical variables were transformed using **One-Hot Encoding** to prepare for binary classification.

Exploratory Data Analysis (EDA):

Visualization and SQL queries were performed to detect trends, correlations, and anomalies.

Interactive Visualization:

Tools such as **Folium** and **Plotly Dash** were used to create **interactive dashboards and geospatial maps** for better insights.

Predictive Modeling:

Several machine learning models were developed, trained, and fine-tuned — including **Logistic Regression**, **Support Vector Machines (SVM)**, **Decision Trees**, and **K-Nearest Neighbors (KNN)** — to predict landing outcomes.

Data Collection

The data used in this project were obtained from two main sources: the **SpaceX public API** and a **static Wikipedia page**.

1. SpaceX API

Launch data are accessible via the SpaceX REST API:

<https://api.spacexdata.com/v4/launches/past>

For reproducibility, a static JSON dataset was used:

https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.json

The dataset includes **flight number, launch date, booster version, payload mass, orbit type, launch site, and outcome**.

After filtering for **Falcon 9** launches, missing payload values were imputed with their mean, and the final dataset was exported as a **CSV file** for further analysis.

2. Web Scraping (Wikipedia)

Additional launch data were collected via **web scraping** from the static Wikipedia page:

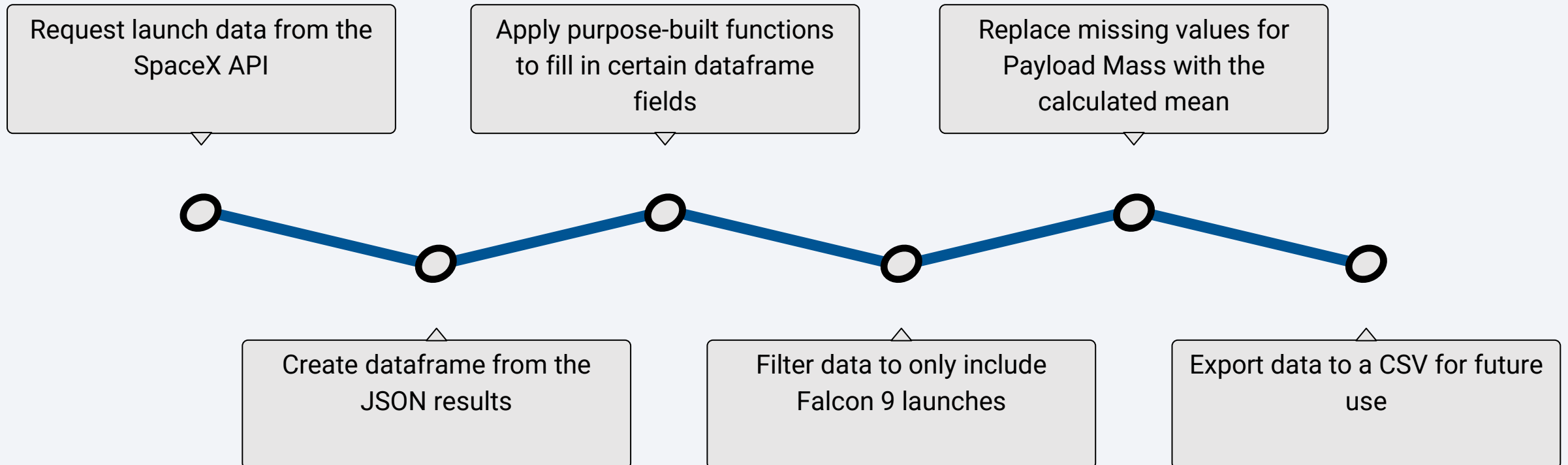
https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches

Using **BeautifulSoup**, the HTML content was parsed to extract key fields such as flight number, payload, payload mass, and launch site.

The extracted information was structured into a **DataFrame** and exported as a **CSV** file for use in subsequent analyses.

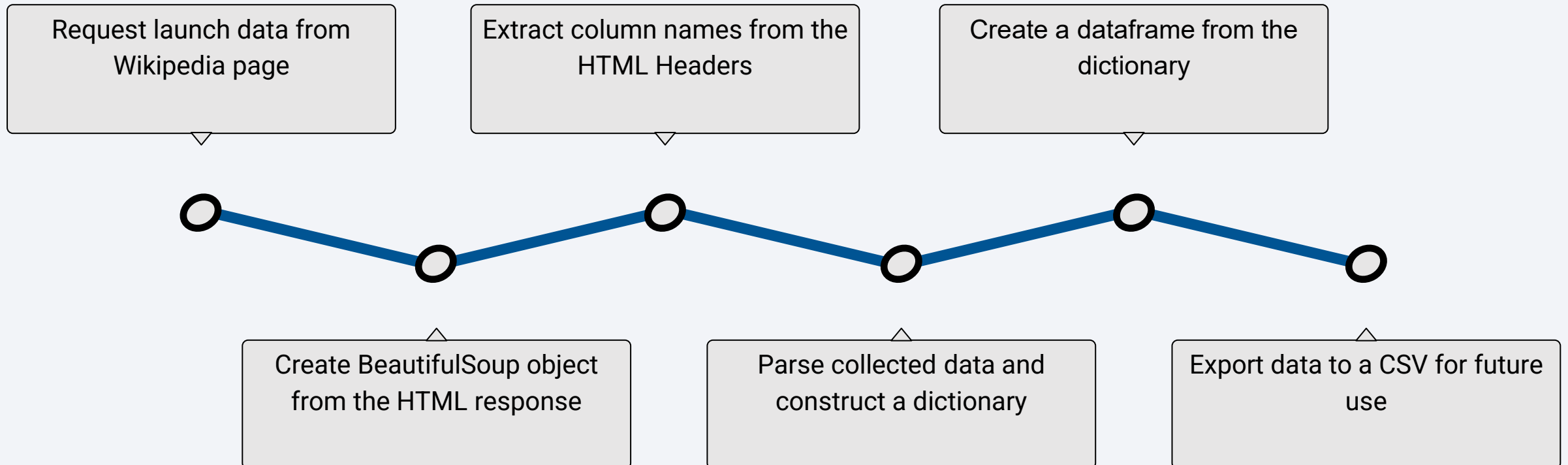
Data Collection – SpaceX API

- REST calls from the SpaceX API was used to collect data including Flight Number, Date, Booster Version, Payload Mass, Orbit Type, Launch Site, Outcome, etc.



Data Collection - Scraping

- Web scraping was performed using the Wikipedia page on SpaceX launches to obtain data including Flight Number, Launch Site, Payload, Payload Mass, etc.



Data Wrangling

The **data wrangling** phase consisted of the following tasks:

Calculating the total number of launches per site

Computing the frequency of each orbit type

Determining mission outcomes for each orbit category

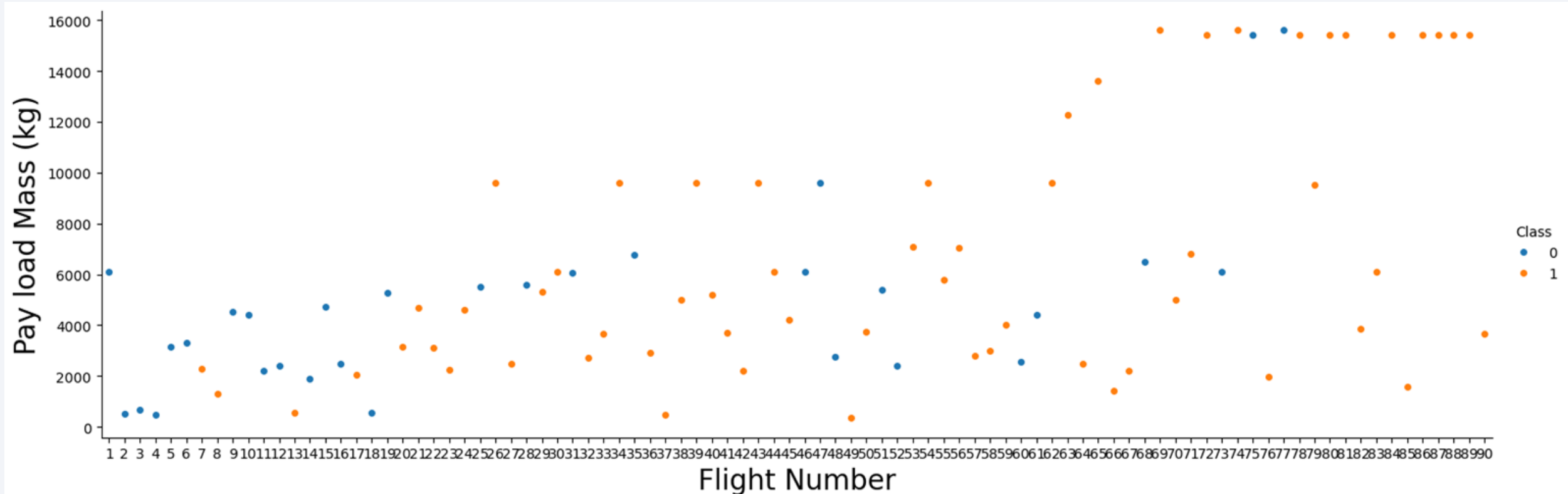
Creating a **binary landing outcome label** from the original outcome field

Exporting the processed dataset for modeling

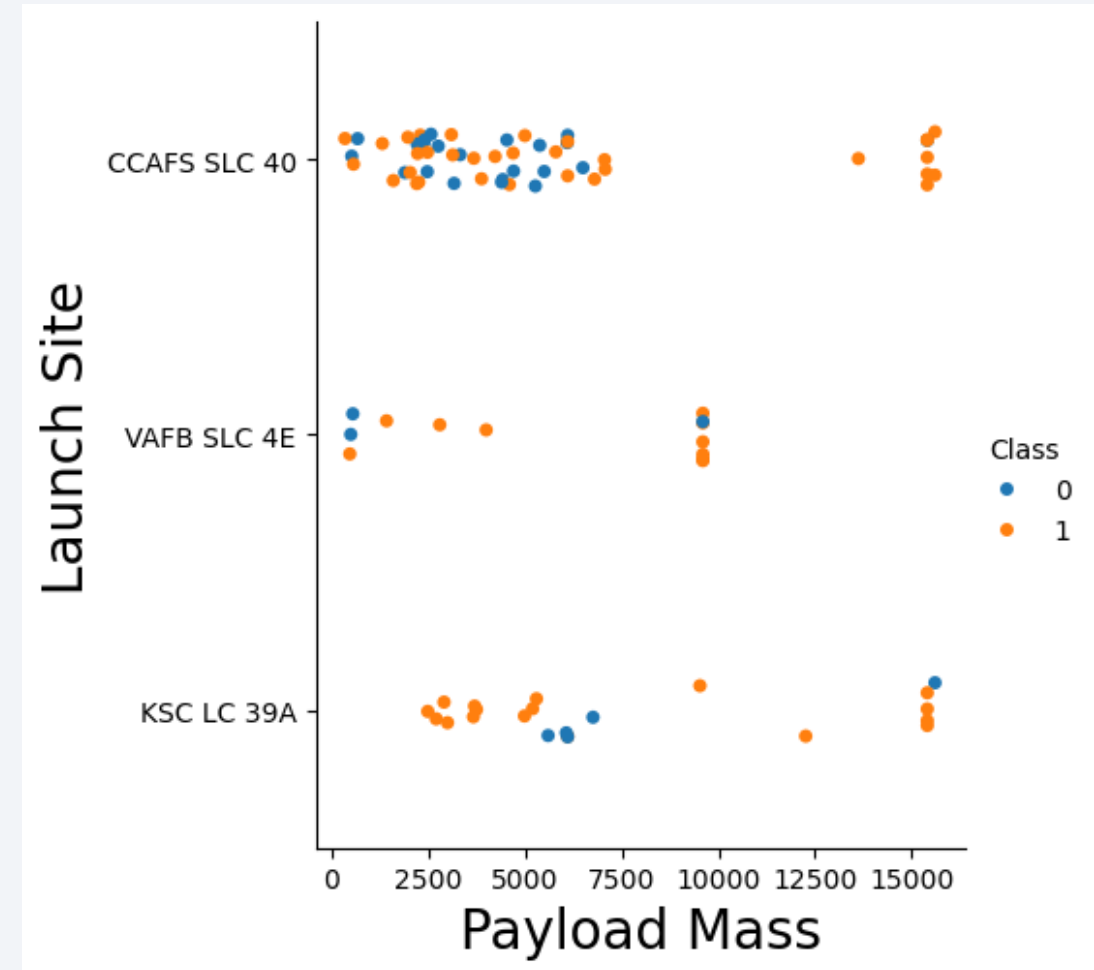
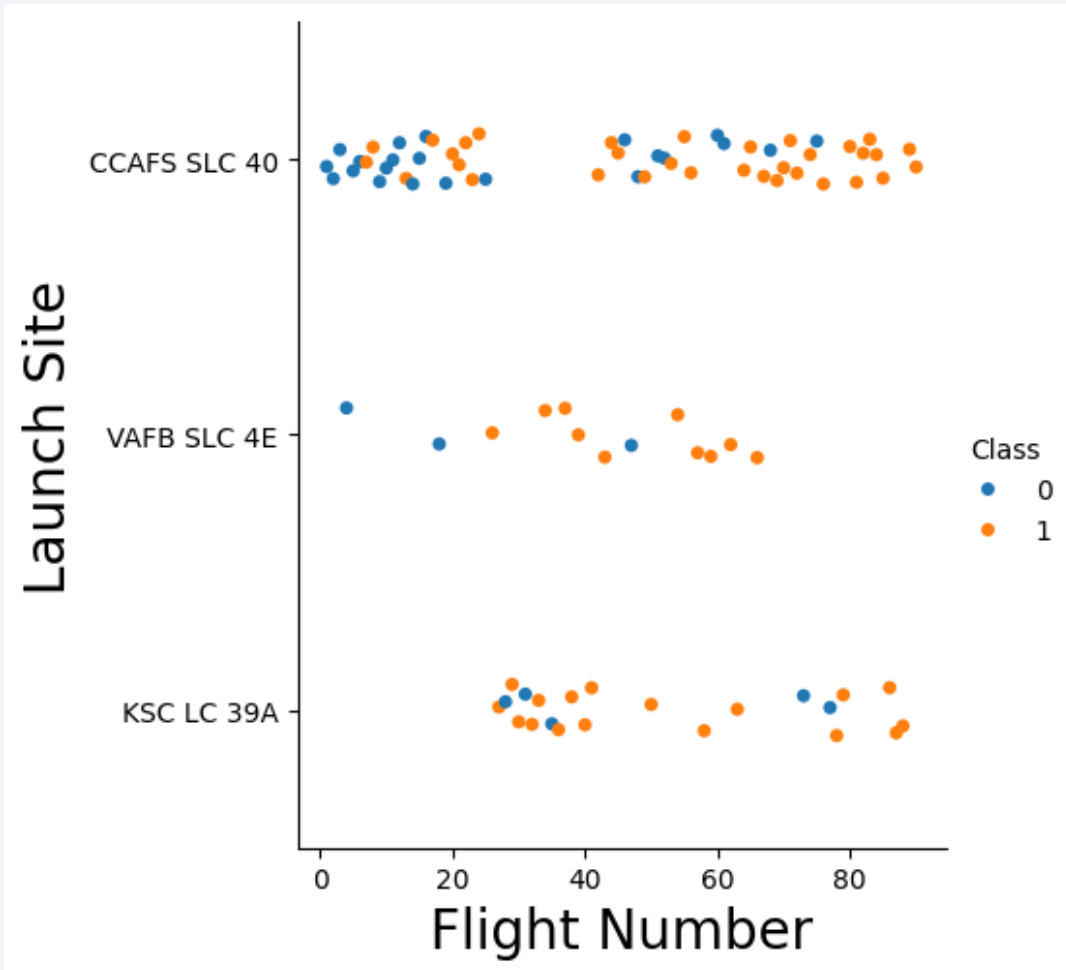
EDA with Data Visualization

- Exploratory data analysis was performed, and plots, including scatter plots, bar charts, and line plots were created to explore relationships between flight number, payload mass, launch site, orbit type, year, and success rate.
- Example plots are shown on the following slides.

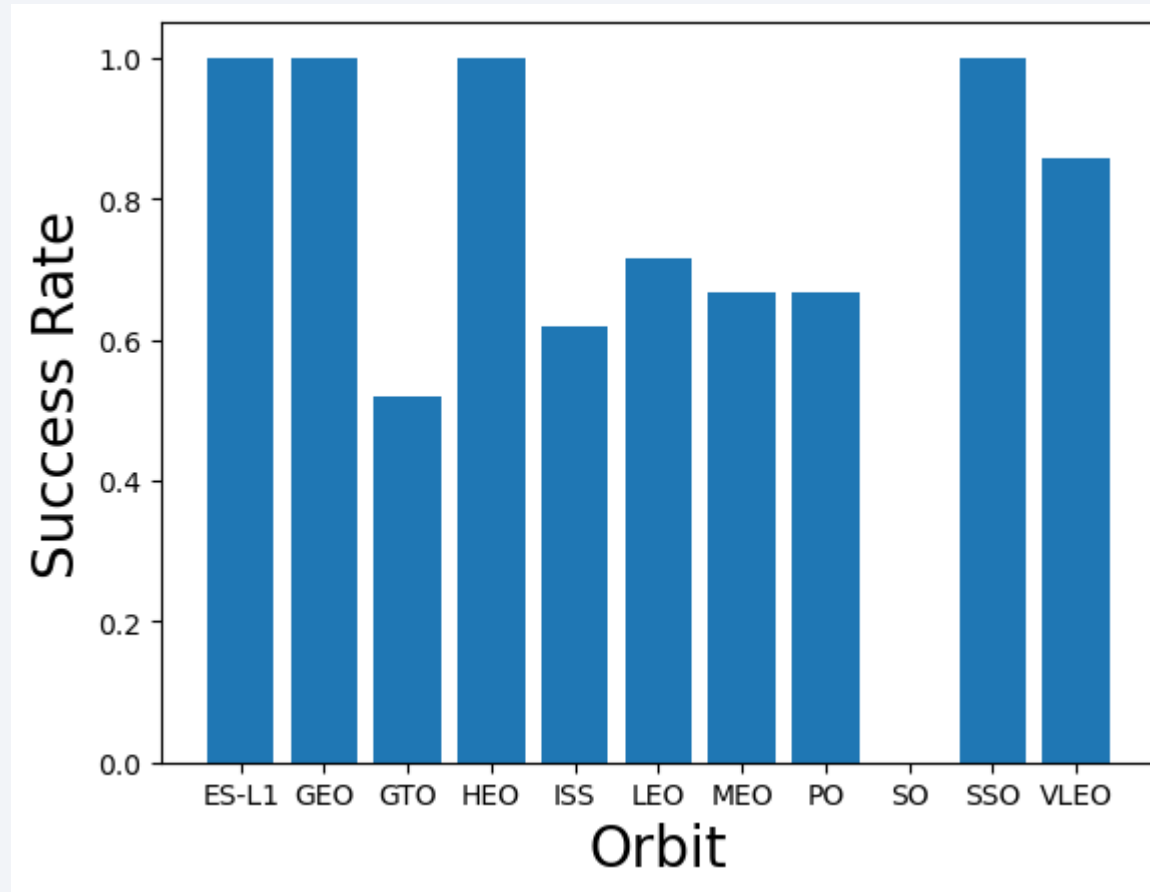
EDA with Data Visualization



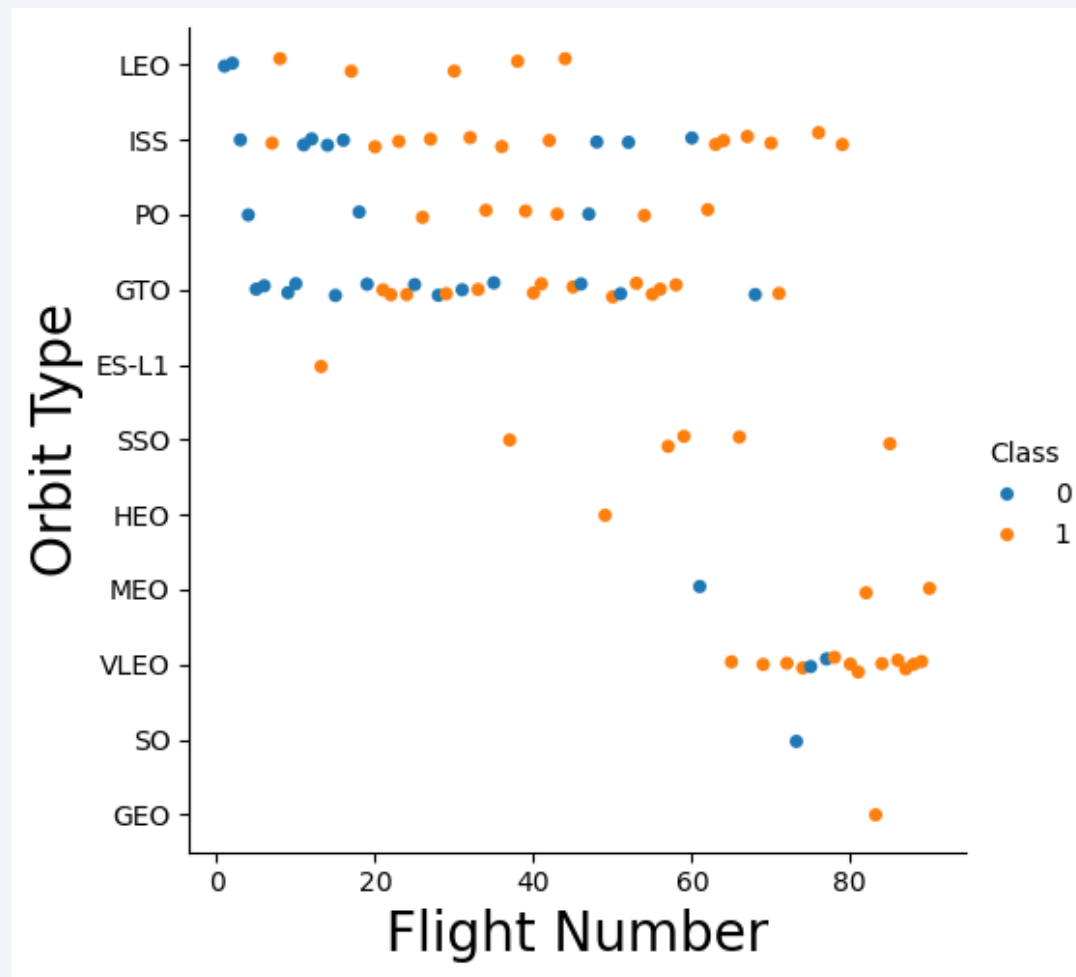
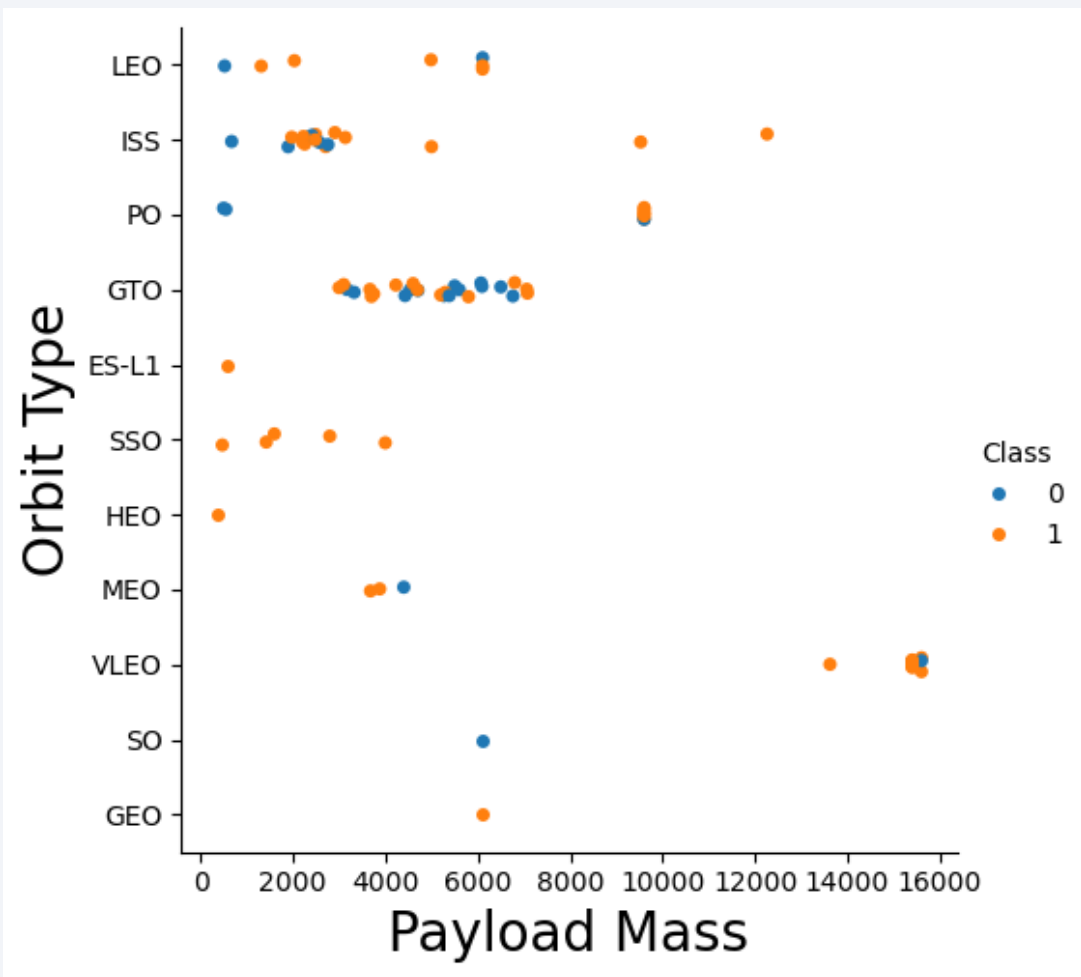
EDA with Data Visualization



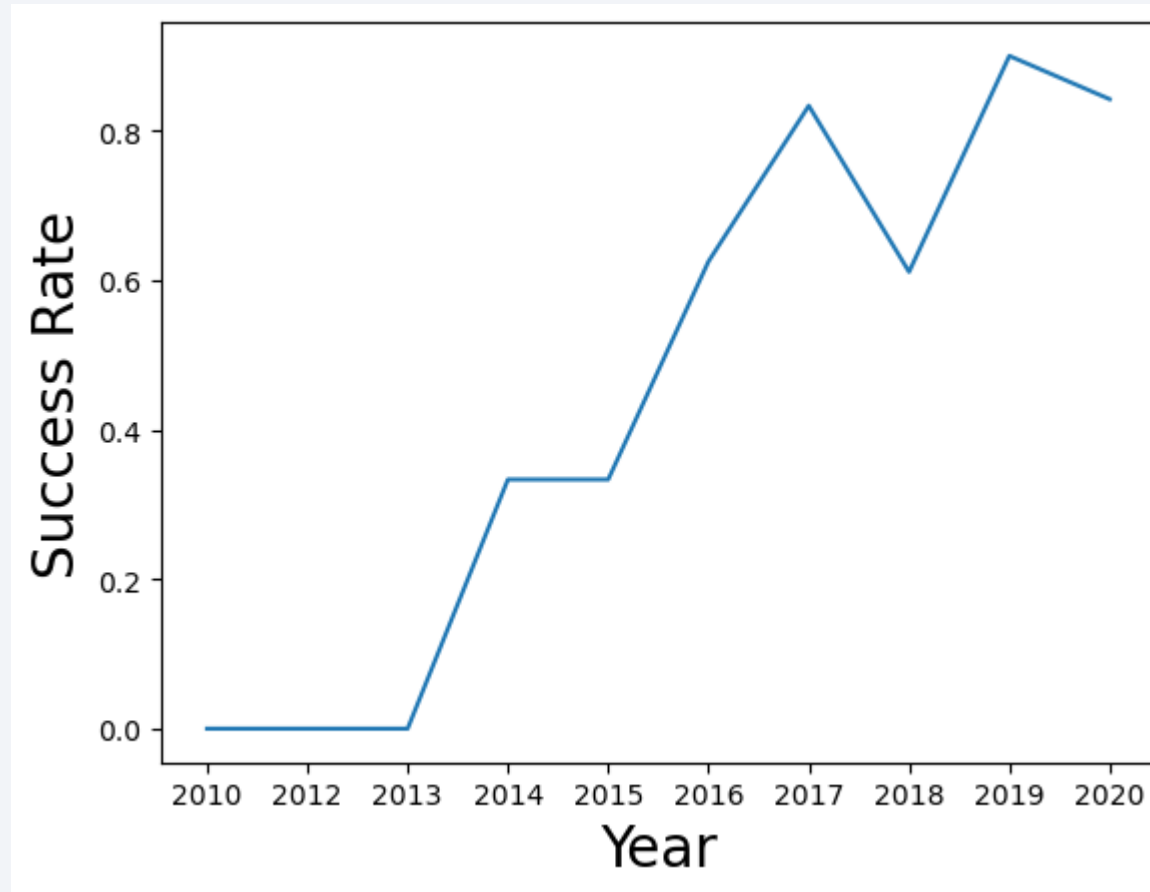
EDA with Data Visualization



EDA with Data Visualization



EDA with Data Visualization



EDA with SQL

We executed SQL analyses to:

Identify all **unique launch sites** in the dataset.

Retrieve **five** records where the **launch site** begins with “**CCA**”.

Compute the **total payload mass** carried by boosters on **NASA (CRS)** missions.

Calculate the **average payload mass** for **booster version F9 v1.1**.

Find the **first date** on which a **successful ground-pad landing** was achieved.

List the **booster names** that **succeeded on a drone ship** with **payload mass between 4,000 and 6,000 kg**.

Count **successful vs. failed mission outcomes**.

Return the **booster versions** that carried the **maximum payload mass** (via a subquery).

For **2015**, show **month names**, **failed drone-ship landing outcomes**, **booster versions**, and **launch sites**.

Rank landing outcomes by their **occurrence count** between **2010-06-04** and **2017-03-20** (descending).

Build an Interactive Map with Folium

- An interactive folium map was created to show:
 - A circle marker at each launch site for the successful and failed launches
 - A polyline showing the distance to the nearest coastline

Build a Dashboard with Plotly Dash

- An interactive dashboard was created to show the following:
 - A pie chart of total successful launches by site
 - Selectable pie charts for successful launched from each site
 - A plot of payload mass vs.outcome for all sites with color coding by booster version
 - A plot of payload mass vs.outcome for a selected site with color coding by booster version
- Screenshots of the dashboard are shown on the following slides

Build a Dashboard with Plotly Dash

SpaceX Launch Records Dashboard

All Sites

✕ ▼

Total Successful Launches by Site

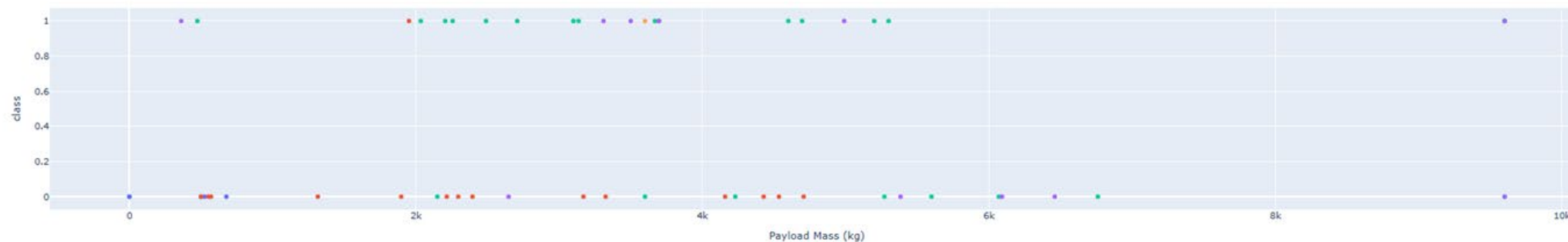


■ KSC LC-39A
■ CCAFS LC-40
■ VAFB SLC-4E
■ CCAFS SLC-40

Payload range (Kg):



Payload vs. Outcome for All Sites



Booster Version Category
● v1.0
● v1.1
● FT
● B4
● B5

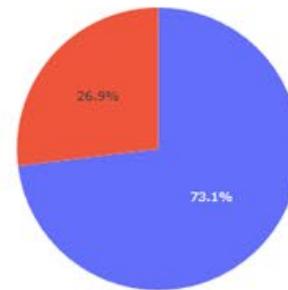
Build a Dashboard with Plotly Dash

SpaceX Launch Records Dashboard

CCAFS LC-40

×

Success vs Failure for site CCAFS LC-40



■ Failed Launch
■ Successful Launch

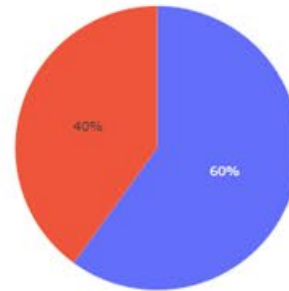
Build a Dashboard with Plotly Dash

SpaceX Launch Records Dashboard

VAFB SLC-4E

×

Success vs Failure for site VAFB SLC-4E



■ Failed Launch
■ Successful Launch

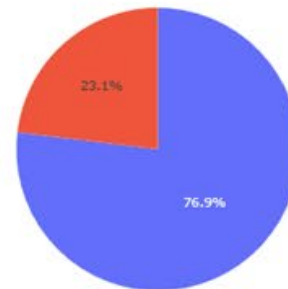
Build a Dashboard with Plotly Dash

SpaceX Launch Records Dashboard

KSC LC-39A

×

Success vs Failure for site KSC LC-39A



■ Successful Launch
■ Failed Launch

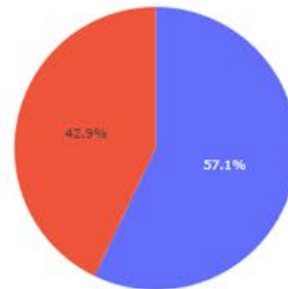
Build a Dashboard with Plotly Dash

SpaceX Launch Records Dashboard

CCAFS SLC-40

×

Success vs Failure for site CCAFS SLC-40



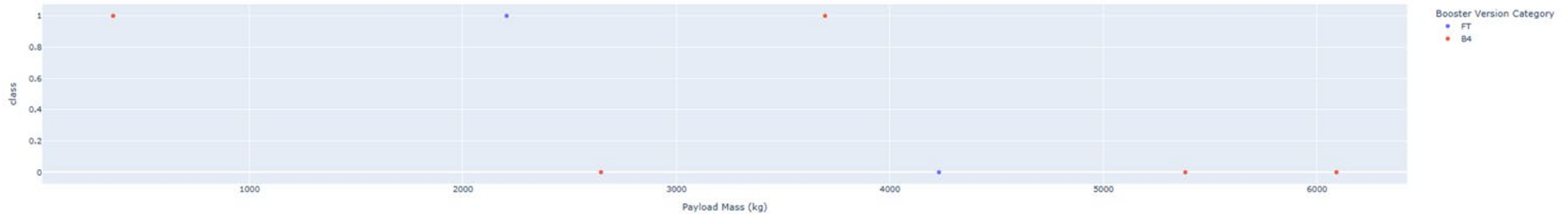
■ Failed Launch
■ Successful Launch

Build a Dashboard with Plotly Dash

Payload range (Kg):



Payload vs. Outcome for site CCAFS SLC-40

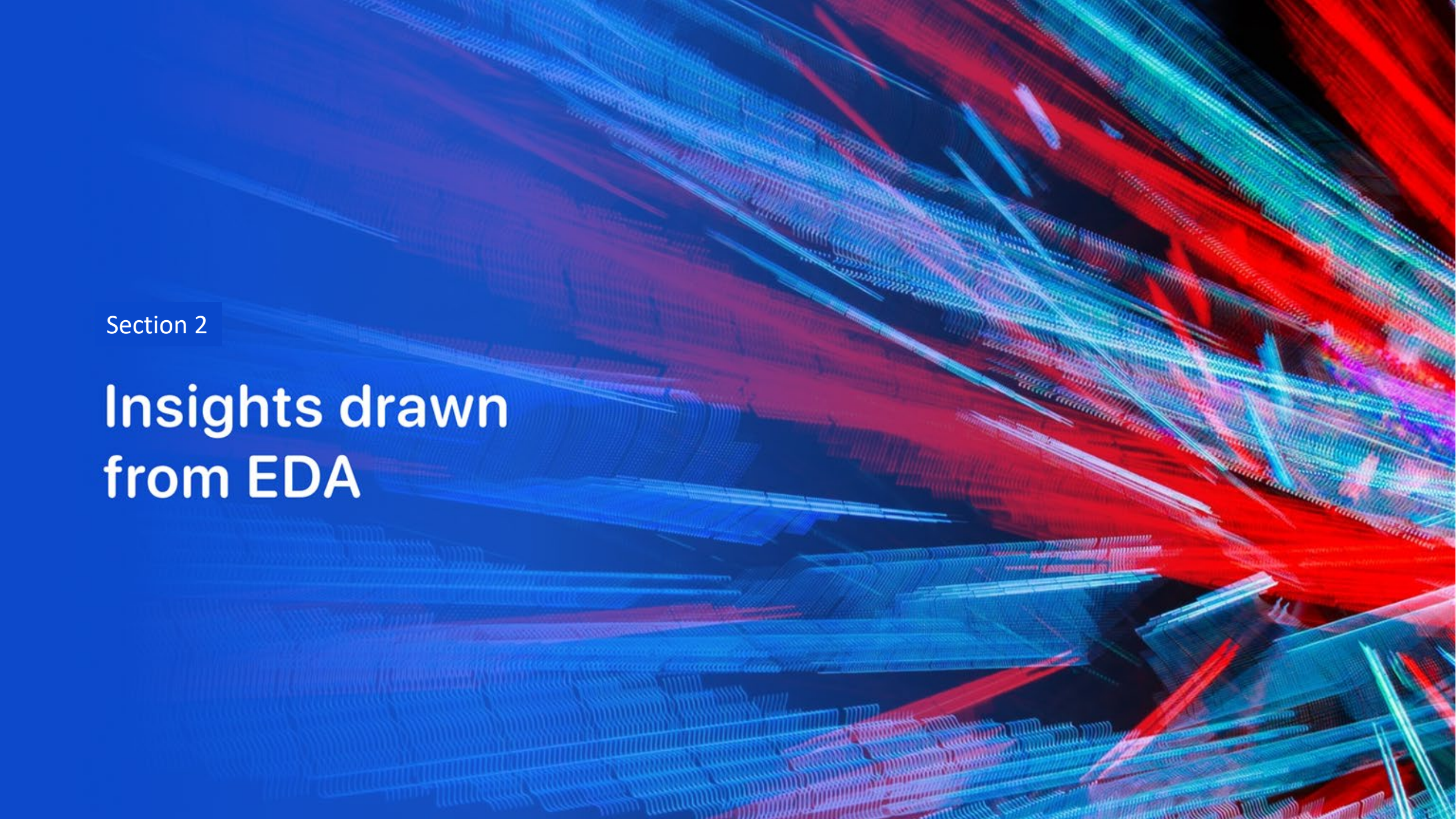


Predictive Analysis (Classification)

- Models were developed using the following methods:
 - Logistic Regression
 - Support Vector Machine
 - Decision Tree
 - K-Nearest Neighbors
- For each model, tuned parameters were determined, and the model was evaluated using a confusion matrix and the method score

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

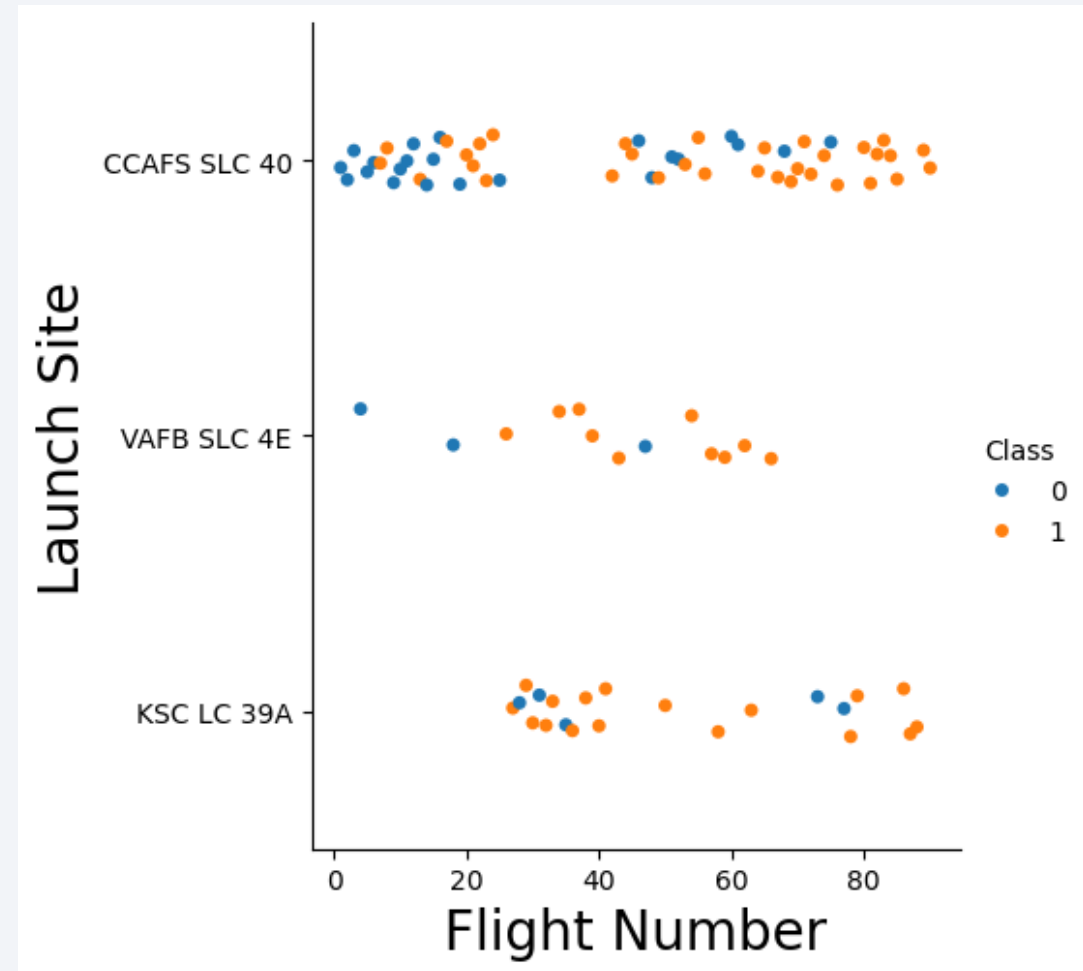


Section 2

Insights drawn from EDA

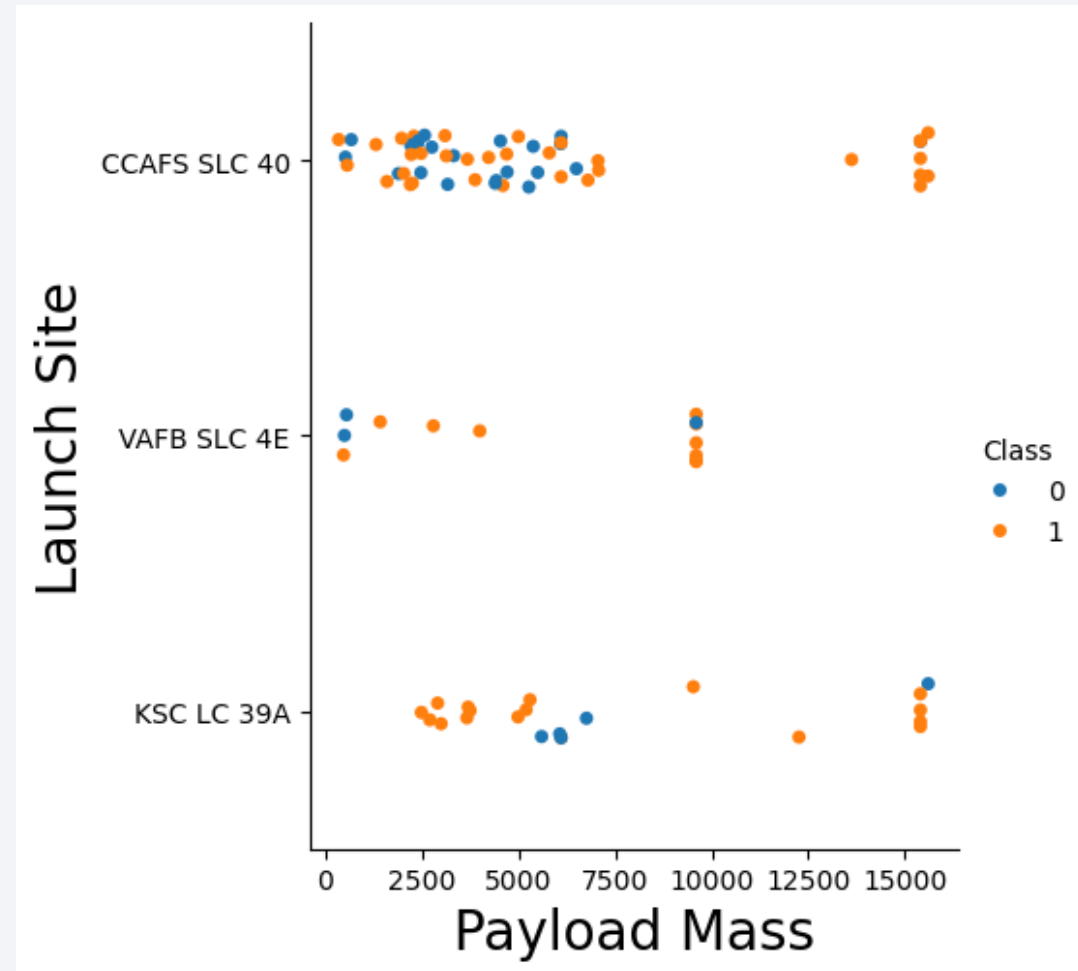
Flight Number vs. Launch Site

- The plot shows increasing success rates with the later flights.
- The plot shows that more launches occurred at the CCAFS SLC 40 site than at either of the other two sites
- The plot does not show a clear trend in where launches occur as a function of flight number



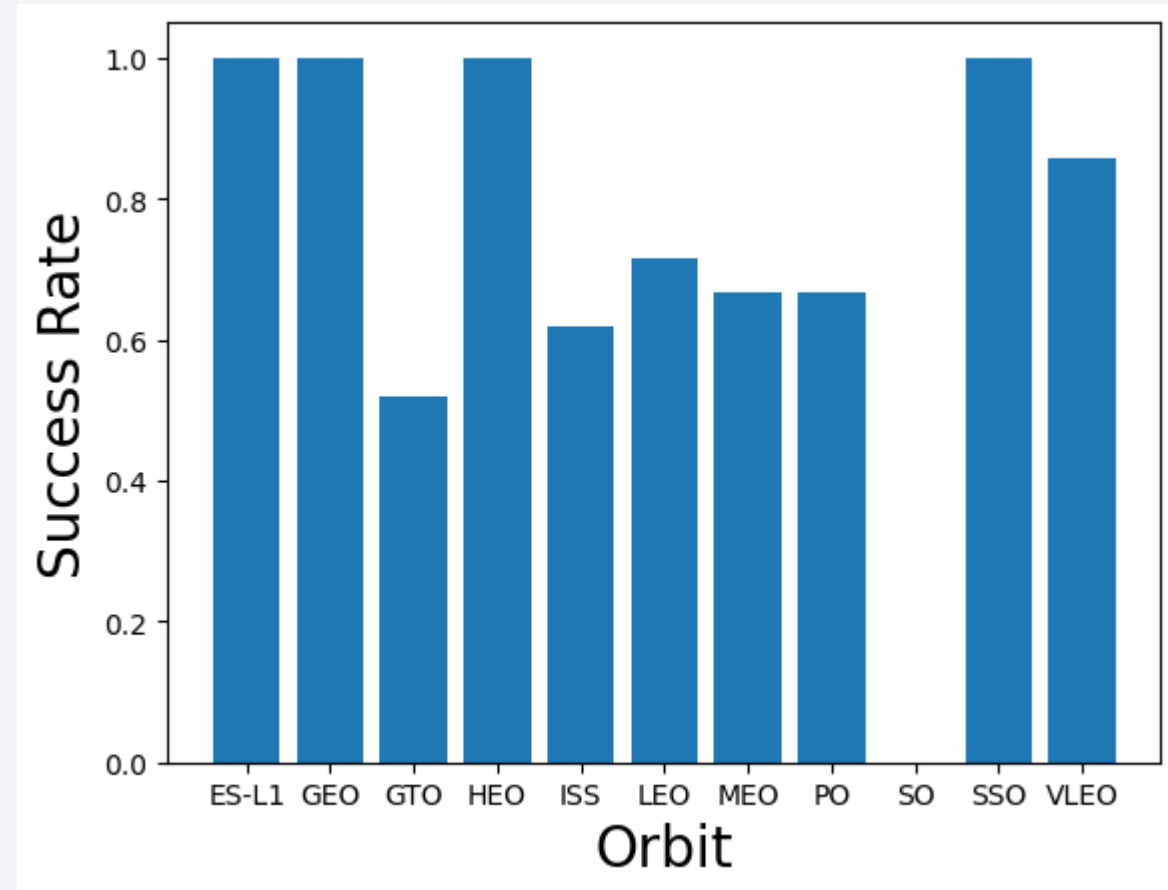
Payload vs. Launch Site

- The plot shows the range of payload masses that were handled at each launch site
- The highest payload masses appear to have greater success rates
- The highest payload masses were only launched from the CCAFS SLC 40 and KSC LC 39A sites



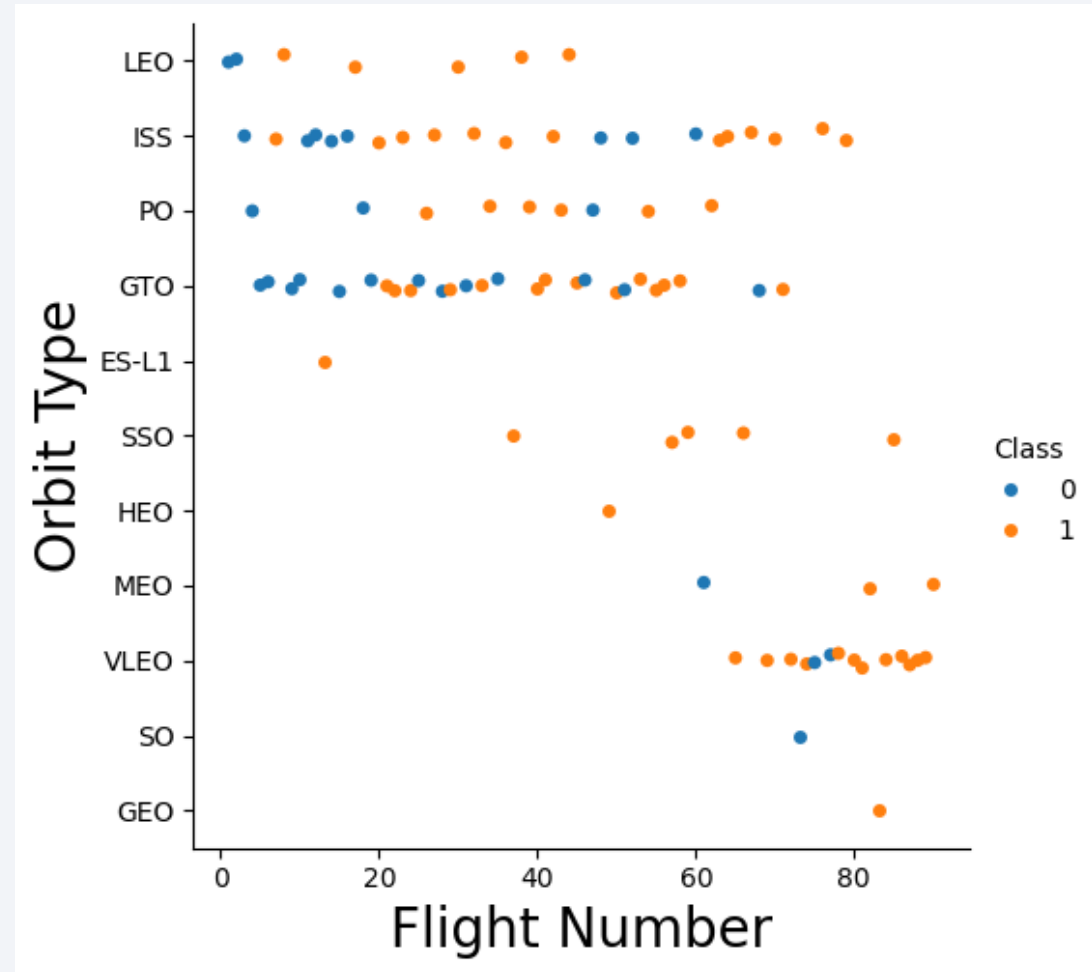
Success Rate vs. Orbit Type

- Several orbit types have 100% success rates:
 - ES-L1, GEO, HEO, and SSO
- One orbit type has a 0% success rate
 - SO
- The success rate for other sites ranges from about 50% to about 85%



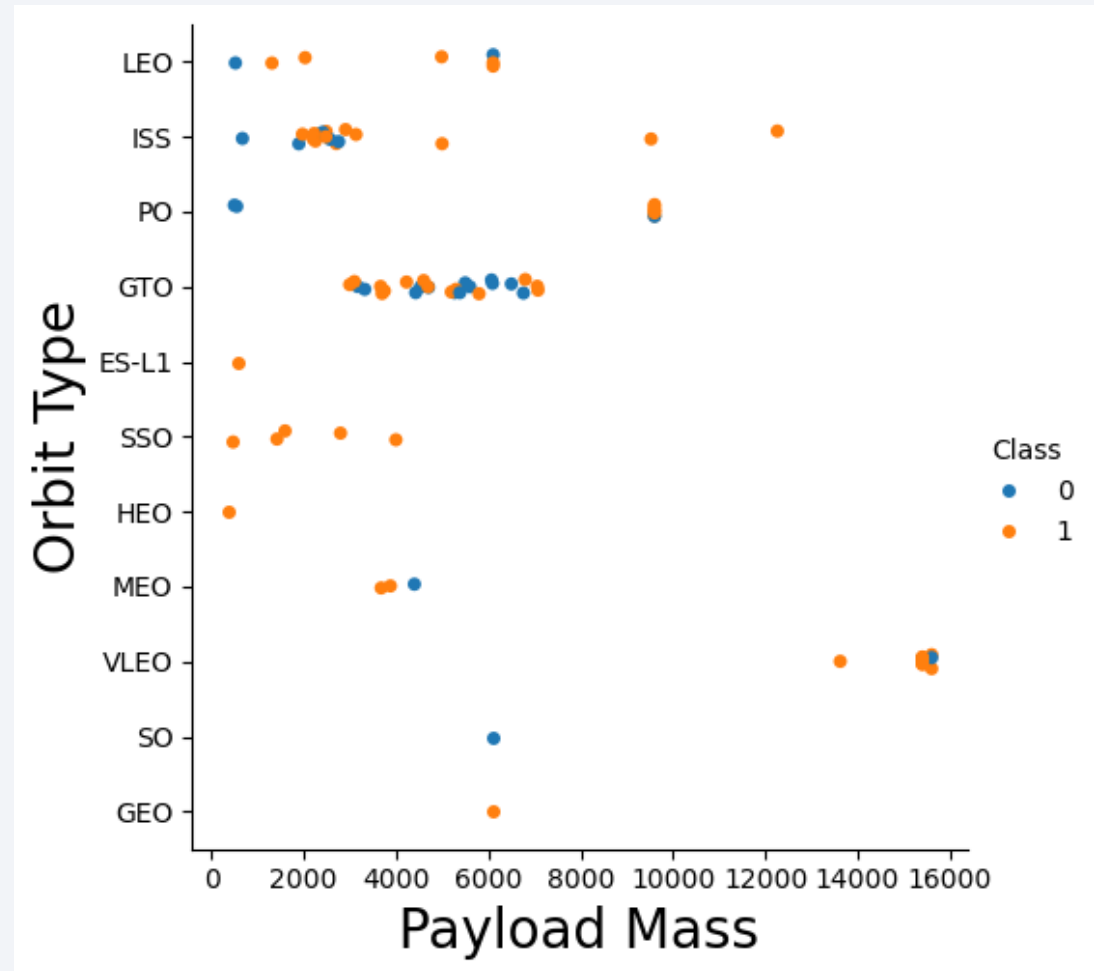
Flight Number vs. Orbit Type

- The chart shows a trend in changes over time with the types of orbits that are being launched.
- LEO, ISS, PO, GTO, and ES-L1 launches have been conducted since fairly early.
- More recently, SSO, HEO, MEO, VLEO, SO, and GEO launches have been conducted.
- Success rates for each type of launch can also be visualized on the plot



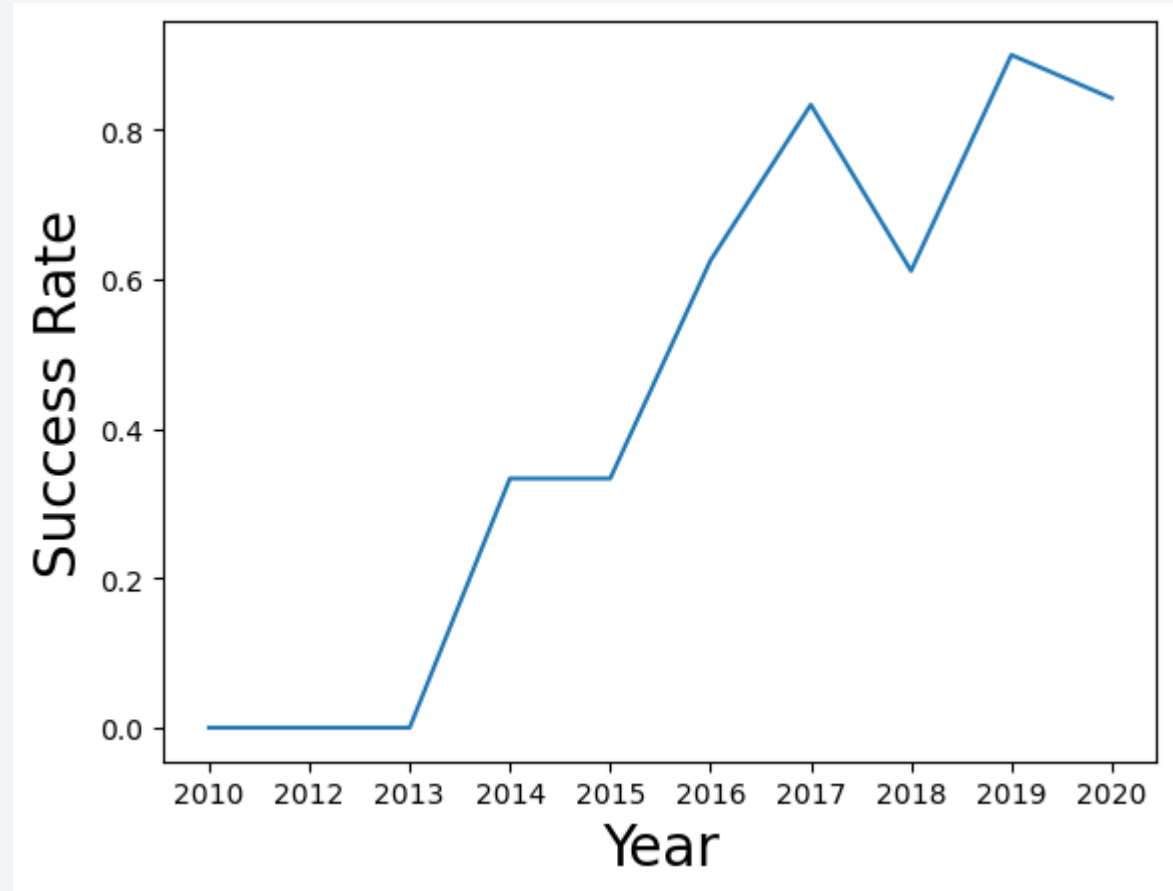
Payload vs. Orbit Type

- The orbit type as a function of payload mass is shown on the plot.
- The heaviest payloads are handled only by the VLEO orbit type.
- The success rate as a function of orbit type and payload mass can also be seen



Launch Success Yearly Trend

- The plot shows a fairly steady increase in success rate
- Success rate in the early years was 0%
- Success in the most recent years was above 80%



All Launch Site Names

- Unique launch site names were found with the following query:

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTBL
```

- Results:

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- The sites beginning with CCA were found with this query:

```
%sql SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

- Result:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Total payload mass was found with this query:

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass  
FROM SPACEXTBL WHERE Customer = 'NASA (CRS)';
```

- Result: 45596

Average Payload Mass by F9 v1.1

- Average payload mass carried by booster version F9 V1.1 was found with this query:

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) AS Avg_Payload_Mass  
FROM SPACEXTBL WHERE Booster_Version = 'F9 v1.1';
```

- Result: 2928.4

First Successful Ground Landing Date

- The date of the first successful landing outcome on ground pad was found with this query:

```
%sql SELECT MIN(Date) AS First_Successful_Ground_Pad_Landing  
FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)';
```

- Result: 2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- The names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000 were found with this query:

```
%sql SELECT DISTINCT Landing_Outcome FROM  
SPACEXTBL
```

- Result shown to the right

Landing_Outcome
Failure (parachute)
No attempt
Uncontrolled (ocean)
Controlled (ocean)
Failure (drone ship)
Precluded (drone ship)
Success (ground pad)
Success (drone ship)
Success
Failure
No attempt

Total Number of Successful and Failure Mission Outcomes

- The total number of successful and failure mission outcomes was found with the following query:

```
%sql SELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE  
Landing_Outcome = 'Success (drone ship)' AND  
PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

- Result:

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Boosters Carried Maximum Payload

- The names of the booster which have carried the maximum payload mass was found with this query:

```
%sql SELECT Booster_Version FROM  
SPACEXTBL WHERE PAYLOAD_MASS__KG_  
= (SELECT MAX(PAYLOAD_MASS__KG_)  
FROM SPACEXTBL);
```

- The results are shown to the right

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

2015 Launch Records

- The failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015 were found with this query:

```
%sql SELECT substr(Date, 6, 2) AS Month,Landing_Outcome,  
Booster_Version, Launch_Site FROM SPACEXTBL WHERE substr(Date,  
1, 4) = '2015' AND Landing_Outcome = 'Failure (drone ship)';
```

- Result:

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- The count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, were ranked in descending order using this query:

```
%sql SELECT Landing_Outcome, COUNT(*)  
AS Outcome_Count FROM SPACEXTBL  
WHERE Date BETWEEN '2010-06-04' AND  
'2017-03-20' GROUP BY Landing_Outcome  
ORDER BY Outcome_Count DESC;
```

- Result shown to the right

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

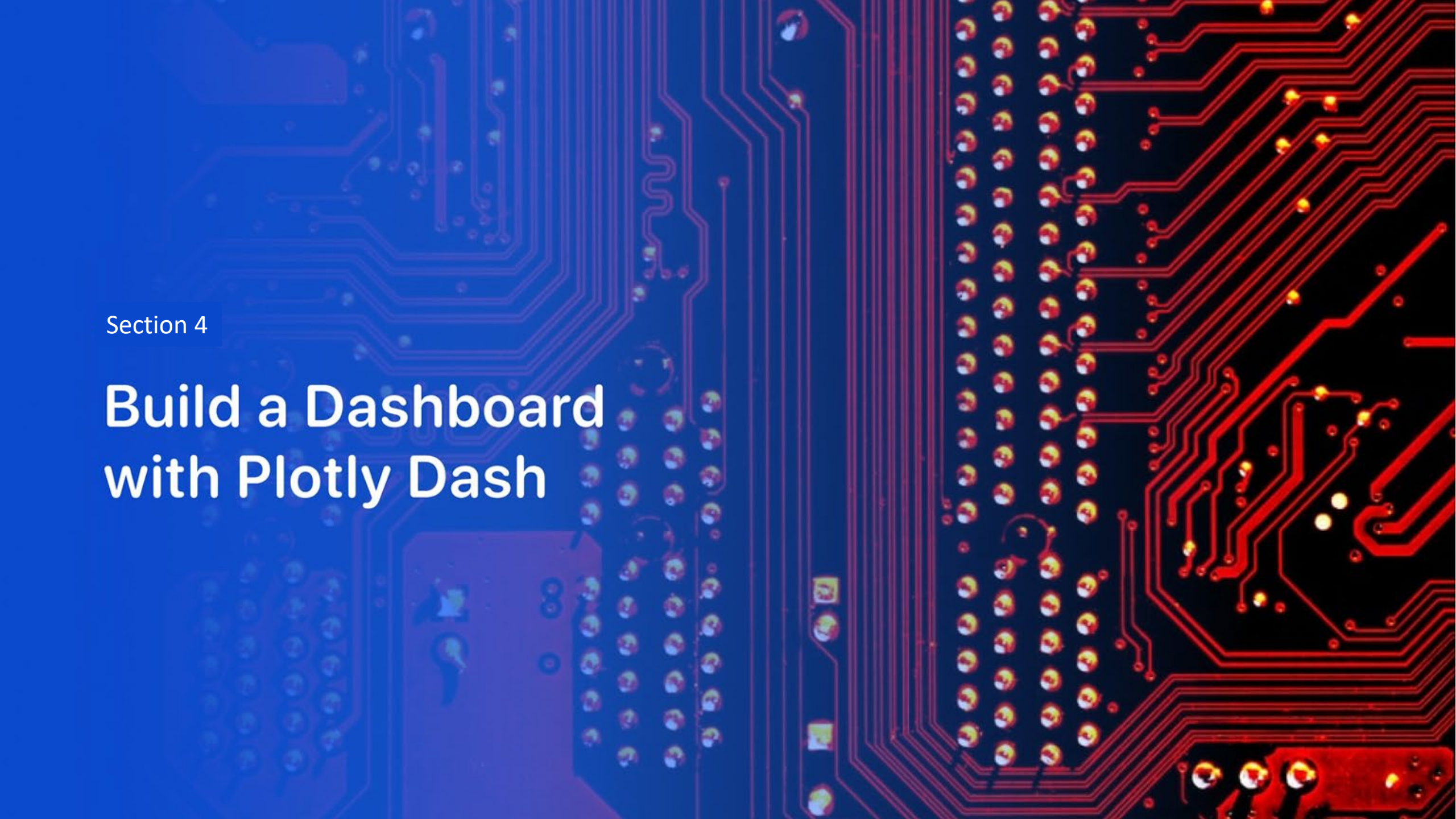
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and clouds. The lights are concentrated in the lower right portion of the image, while the upper left shows a clear blue sky.

Section 3

Launch Sites Proximities Analysis

Folium map screenshots

- Unfortunately, screenshots for the folium map were not obtained.

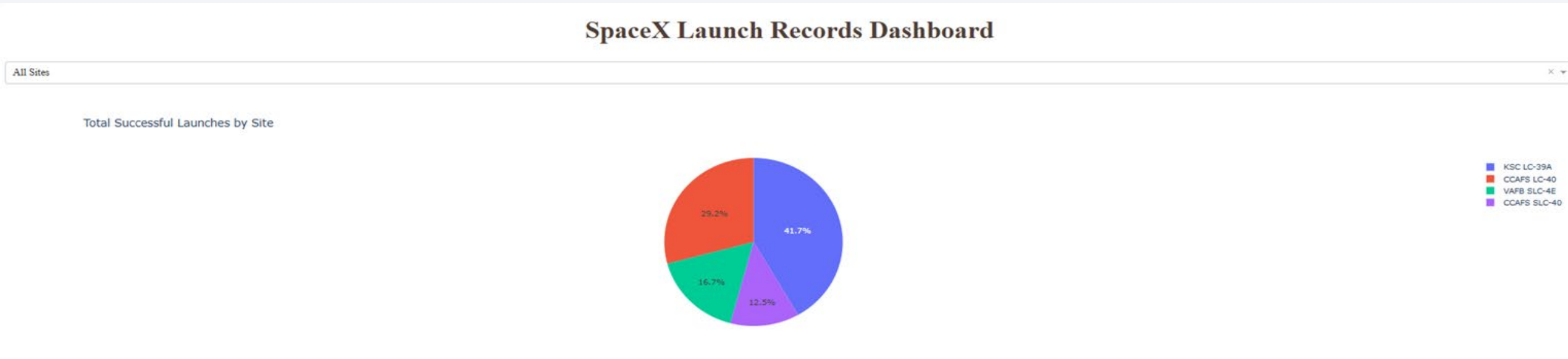


Section 4

Build a Dashboard with Plotly Dash

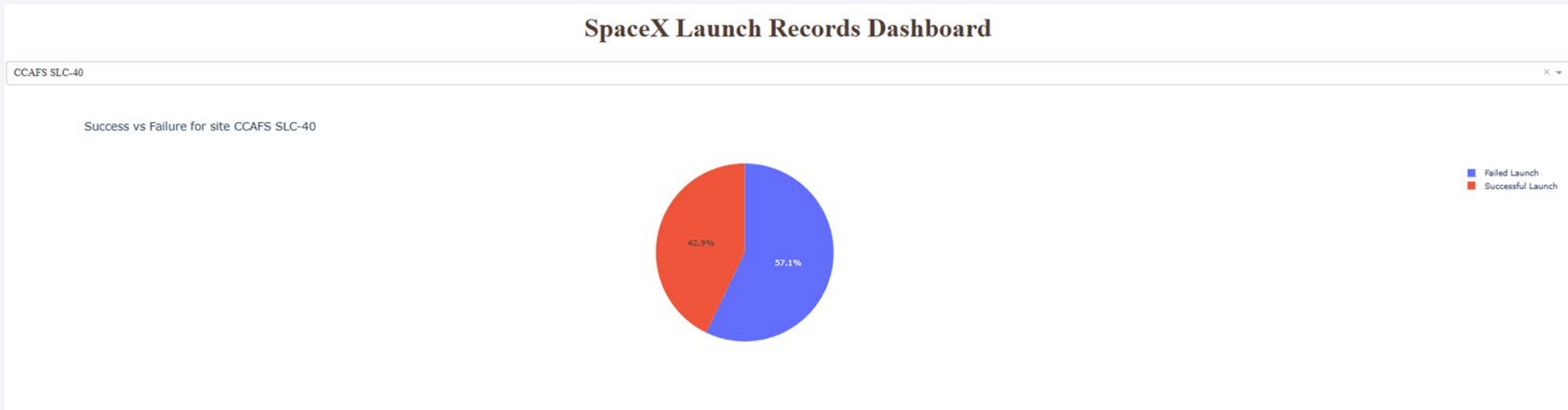
Dashboard Screenshot: Launch Success Count for All Sites

- The plot shows that the site with the most successful launches (41.7% of all successful launches) was KSC LC-39A
- The site with the fewest successful launches was CCAFS SLC-40



Dashboard Screenshot: Site with Highest Success Ratio

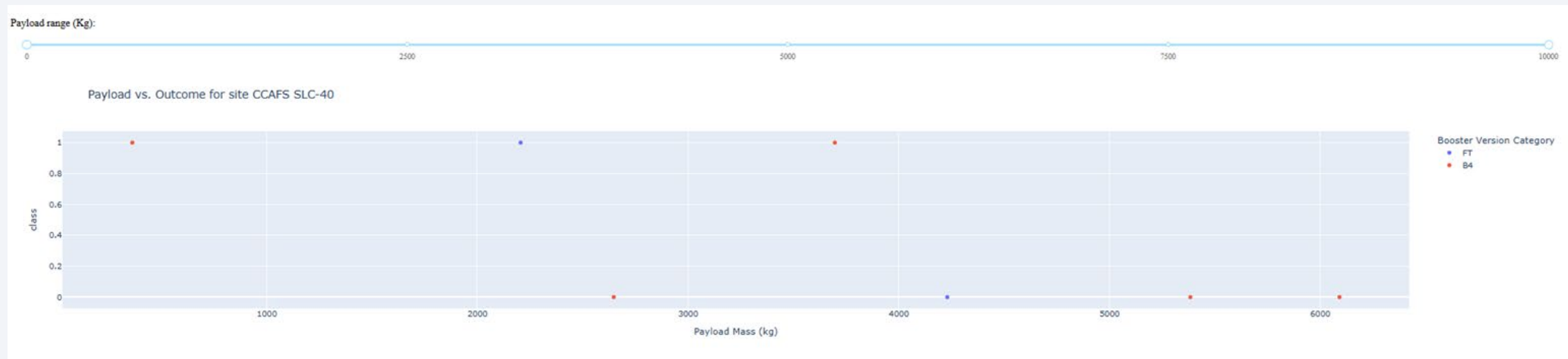
- The site with the highest success rate is CCAFS SLC-40, with a success rate of 42.9%



Dashboard Screenshot:

Payload vs. Launch Outcome scatter plot for CCAFS SLC-40

- The plot shows the payload vs. outcome for CCAFS SLC-40 for the full range of payload masses
- Payload range can be selected with the slider
- A screenshot of the plot for all sites was not taken



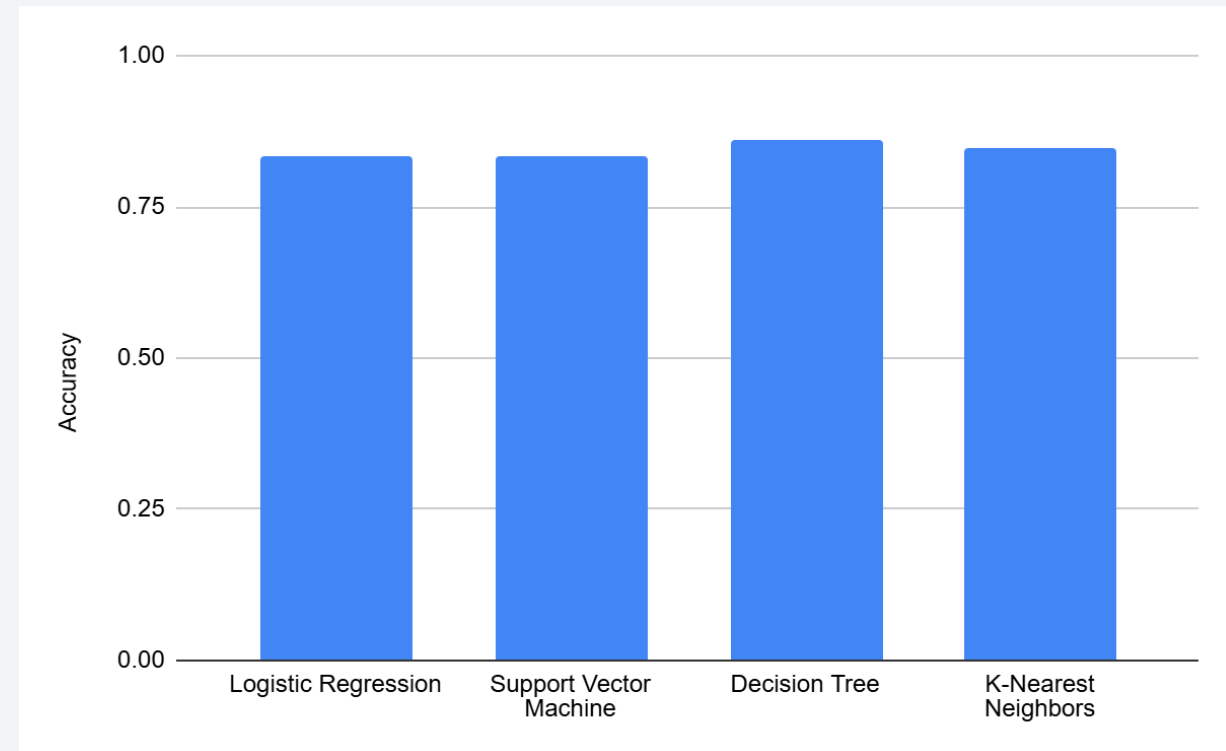


Section 5

Predictive Analysis (Classification)

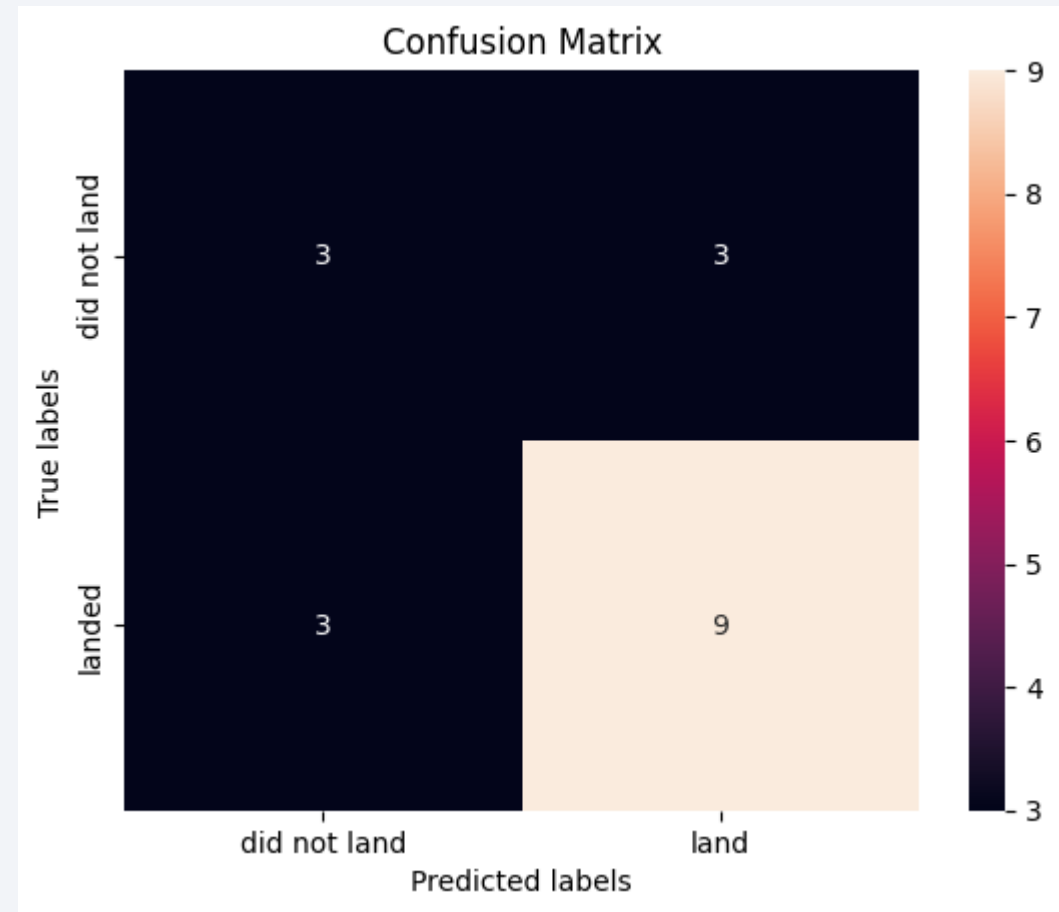
Classification Accuracy

- The model accuracy for each tuned model is shown to the right.
- The Decision Tree model has the highest accuracy.



Confusion Matrix

- The confusion matrix for the Decision Tree model is shown to the right.
- The matrix shows relatively low errors for both Type 1 and Type 2 errors with 3 each.
- The model successfully predicts the correct outcome in 12 of the cases for the test set (9 landed and 3 did not land)



Conclusions

- The Decision Tree Model performs best for this dataset
- Launches with higher payload mass show greater success rates
- Most launches are from sites close to the equator and close to the coastline
- Success rates have increased over time

Thank you!

